Lesson 37 material: Final Project: Your Research Proposal in APA Style

2.9: Overview of the Scientific Method (Summary)

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Changes: formatted for print; exercise answers are in the study guide.

Key Takeaways

- Research in psychology can be described by a simple cyclical model. A research question based on the research literature leads to an empirical study, the results of which are published and become part of the research literature.
- The research literature in psychology is all the published research in psychology, consisting primarily of articles in professional journals and scholarly books.
- Early in the research process, it is important to conduct a review of the research literature on your topic to refine your research question, identify appropriate research methods, place your question in the context of other research, and prepare to write an effective research report.
- There are several strategies for finding previous research on your topic. Among the best is using PsycINFO, a computer database that catalogs millions of articles, books, and book chapters in psychology and related fields.
- Research questions expressed in terms of variables and relationships between variables can be suggested by other researchers or generated by asking a series of more general questions about the behavior or psychological characteristic of interest.
- It is important to evaluate how interesting a research question is before designing a study and collecting data to answer it. Factors that affect interestingness are the extent to which the answer is in doubt, whether it fills a gap in the research literature, and whether it has important practical implications.
- It is also important to evaluate how feasible a research question will be to answer. Factors that affect feasibility include time, money, technical knowledge and skill, and access to special equipment and research participants.
- A theory is broad in nature and explains larger bodies of data. A hypothesis is more specific and makes a prediction about the outcome of a particular study.
- Working with theories is not “icing on the cake.” It is a basic ingredient of psychological research.
- Like other scientists, psychologists use the hypothetico-deductive method. They construct theories to explain or interpret phenomena (or work with existing theories), derive hypotheses from their theories, test the hypotheses, and then reevaluate the theories in light of the new results.
- Variables vary across people or situations and may be quantitative (e.g., age) or categorical (e.g., course subject).
- A sample is a small subset of a larger population that is selected to participate in the research study. There are many different ways of sampling participants including convenience sampling and simple random sampling.
- Experimental research involves manipulating an independent variable to observe the effects on a measured dependent variable while non-experimental research involves measuring variables as they naturally occur (i.e., without manipulating anything).
- Research can be conducted in the field or the lab. Laboratory experiments tend to have high internal validity (allowing us to make strong causal conclusions), while field studies often have more external validity (allowing us to generalize to the real world).
- The mean, median, and mode are measures of central tendency used to describe the typical, average, or center scores in a distribution. The range, standard deviation, and variance are measures of how dispersed or spread apart the scores are. Measures of central tendency and dispersion are important descriptive statistics.
- Inferential statistics allow researchers to determine whether their findings are statistically significant, that is, whether they are unlikely to be due to chance alone and therefore are likely to represent a real effect in the population.
- Since statistics are probabilistic in nature we never know if our conclusions are correct. We can make type I errors (concluding an effect is real when it is not) or type II errors (concluding there is no effect when there actually is a real effect in the population).
- Theories can be supported by not proved. Similarly, disconfirming a hypothesis does not necessarily mean that theory has been disproved.
- The final step of the research process involves reporting results at scientific conferences, in journal articles, and/or in books.

References

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Exercises

- Practice: Find a description of an empirical study in a professional journal or in one of the scientific psychology blogs. Then write a brief description of the research in terms of the cyclical model presented here. One or two sentences for each part of the cycle should suffice.
- Watch the following TED Ed video, in which David H. Schwartz provides an introduction to two types of empirical studies along with some methods that scientists use to increase the reliability of their results:

A YouTube element has been excluded from this version of the text. You can view it online here: https://kpu.pressbooks.pub/psychmethods4e/?p=271

  - Reading in print? Go to https://youtu.be/GUpd2HJHUt8 or scan this QR code with your phone:
- Practice: Use the techniques discussed in this section to find 10 journal articles and book chapters on one of the following research ideas: memory for smells, aggressive driving, the causes of narcissistic personality disorder, the functions of the intraparietal sulcus, or prejudice against the physically handicapped.
- Watch the following video clip produced by UBCiSchool about how to read an academic paper (without losing your mind):

A YouTube element has been excluded from this version of the text. You can view it online here: https://kpu.pressbooks.pub/psychmethods4e/?p=271

  - Reading in print? Go to https://youtu.be/SKxm2HF_-k0 or scan this QR code with your phone:
- Practice: Generate three research ideas based on each of the following: informal observations, practical problems, and topics discussed in recent issues of professional journals.
- Practice: Generate an empirical research question about each of the following behaviors or psychological characteristics: long-distance running, getting tattooed, social anxiety, bullying, and memory for early childhood events.
- Practice: Evaluate each of the research questions you generated in Exercise 2 in terms of its interestingness based on the criteria discussed in this section.
- Practice: Find an issue of a journal that publishes short empirical research reports (e.g., *Psychological Science*, *Psychonomic Bulletin and Review*, *Personality and Social Psychology Bulletin*). Pick three studies, and rate each one in terms of how feasible it would be for you to replicate it with the resources available to you right now. Use the following rating scale: (1) You could replicate it essentially as reported. (2) You could replicate it with some simplifications. (3) You could not replicate it. Explain each rating.
- Practice: Find a recent empirical research report in a professional journal. Read the introduction and highlight in different colors descriptions of theories and hypotheses.
- Practice: Using the research article you found in a professional journal identify whether the study was experimental or non-experimental. If it was experimental identify the independent and dependent variables.
- Practice: Using the research article you found in a professional journal identify which descriptive statistics were reported.
- Practice: Describe why theories can be supported but not proved.

Links on the original page

- Web page (kpu.pressbooks.pub): https://kpu.pressbooks.pub/psychmethods4e/?p=271#pb-interactive-content
- Video (YouTube): https://youtu.be/GUpd2HJHUt8
- Video (YouTube): https://youtu.be/SKxm2HF_-k0

11.5: Presenting Your Research (Summary)

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Changes: formatted for print; exercise answers are in the study guide.

Key Takeaways

- APA style is a set of guidelines for writing in psychology. It is the genre of writing that psychologists use to communicate about their research with other researchers and practitioners.
- APA style can be seen as having three levels. There is the organization of a research article, the high-level style that includes writing in a formal and straightforward way, and the low-level style that consists of many specific rules of grammar, spelling, formatting of references, and so on.
- References and reference citations are an important part of APA style. There are specific rules for formatting references and for citing them in the text of an article.
- An APA-style empirical research report consists of several standard sections. The main ones are the abstract, introduction, method, results, discussion, and references.
- The introduction consists of an opening that presents the research question, a literature review that describes previous research on the topic, and a closing that restates the research question and comments on the method. The literature review constitutes an argument for why the current study is worth doing.
- The method section describes the method in enough detail that another researcher could replicate the study. At a minimum, it consists of a participants subsection and a design and procedure subsection.
- The results section describes the results in an organized fashion. Each primary result is presented in terms of statistical results but also explained in words.
- The discussion typically summarizes the study, discusses theoretical and practical implications and limitations of the study, and offers suggestions for further research.
- Research in psychology can be presented in several different formats. In addition to APA-style empirical research reports, there are theoretical and review articles; final manuscripts, including dissertations, theses, and student papers; and talks and posters at professional conferences.
- Talks and posters at professional conferences follow some APA style guidelines but are considerably less detailed than APA-style research reports. Their function is to present new research to interested researchers and facilitate further interaction among researchers.

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Exercises

- Practice: Find a description of a research study in a popular magazine, newspaper, blog, or website. Then identify five specific differences between how that description is written and how it would be written in APA style.
- Practice: Find and correct the errors in the following fictional APA-style references and citations.
  - Walters, F. T., and DeLeon, M. (2010). Relationship Between Intrinsic Motivation and Accuracy of Academic Self-Evaluations Among High School Students. Educational Psychology Quarterly, 23, 234–256.
  - Moore, Lilia S. (2007). Ethics in survey research. In M. Williams & P. L. Lee (eds.), Ethical Issues in Psychology (pp. 120–156), Boston, Psychological Research Press.
  - Vang, C., Dumont, L. S., and Prescott, M. P. found that left-handed people have a stronger preference for abstract art than right-handed people (2006).
  - This result has been replicated several times (Williamson, 1998; Pentecost & Garcia, 2006; Armbruster, 2011)

- Practice: Look through an issue of a general interest professional journal (e.g., *Psychological Science*). Read the opening of the first five articles and rate the effectiveness of each one from 1 (*very ineffective*) to 5 (*very effective*). Write a sentence or two explaining each rating.
- Practice: Find a recent article in a professional journal and identify where the opening, literature review, and closing of the introduction begin and end.
- Practice: Find a recent article in a professional journal and highlight in a different color each of the following elements in the discussion: summary, theoretical implications, practical implications, limitations, and suggestions for future research.
- Discussion: Do an Internet search using search terms such as *psychology* and *poster* to find three examples of posters that have been presented at conferences. Based on information in this chapter, what are the main strengths and main weaknesses of each poster?

Links on the original page

- Web page (apa.org): https://www.apa.org/pi/lgbt/resources/guidelines